

Solution Requirements

Date	29/09/2026
Team ID	SWTID-2026-1297
Project Name	AI FAQ Assistant API
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can register and securely log in using email and password. Passwords are protected using bcrypt and authenticated sessions use JWT.</i>	High
2	Authorization levels	<i>Authorized users can access protected FAQ operations, while unauthorized users cannot access restricted endpoints.</i>	High

3	External Interfaces	<i>The system connects with the Google Gemini API to generate AI-powered FAQ and customer-support responses.</i>	<i>High</i>
4	Transactions processing	<i>The API handles requests for user registration, login, FAQ creation, updating, deletion and search.</i>	<i>High</i>
5	Reporting	<i>The system provides FAQ and API responses through its endpoints; no separate reporting dashboard is currently implemented.</i>	<i>Low</i>
6	Business rules, etc.	<i>FAQ records can be created, updated, deleted and searched according to the application's access and validation rules.</i>	<i>High</i>
7	Compliance to laws or regulations	<i>User credentials and application data should be handled securely and sensitive credentials such as API keys should not be exposed in the source code.</i>	<i>Medium</i>
8	Other	<i>API request validation and error handling are provided through middleware to improve reliability and consistency.</i>	<i>Medium</i>

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	<i>API endpoints should process requests efficiently and return responses without unnecessary delay.</i>	<i>Normal API requests should respond within approximately 2 seconds under normal local testing conditions.</i>
2	Scalability	<i>The system should support increasing numbers of users, FAQ records and API requests.</i>	<i>Can be scaled using cloud deployment, multiple server instances and database optimization.</i>
3	Security & Data Privacy	<i>User passwords are hashed using bcrypt, protected routes use JWT authentication, and sensitive configuration is stored using environment variables.</i>	<i>Passwords must not be stored as plain text and API keys must not be committed to GitHub.</i>
4	Reliability & Availability	<i>The API should handle invalid requests and application errors without crashing the entire server.</i>	<i>Proper validation and error-handling middleware should return appropriate error responses.</i>
5	Usability & Accessibility	<i>API endpoints should have clear request and response structures and be easy to test using an API client.</i>	<i>All required endpoints can be tested using Thunder Client with understandable</i>

			<i>responses.</i>
6	<i>Other</i>	<i>The code should be modular and organized into controllers, routes, models, middleware, services and utility modules.</i>	<i>New features should be addable without significantly changing unrelated modules.</i>